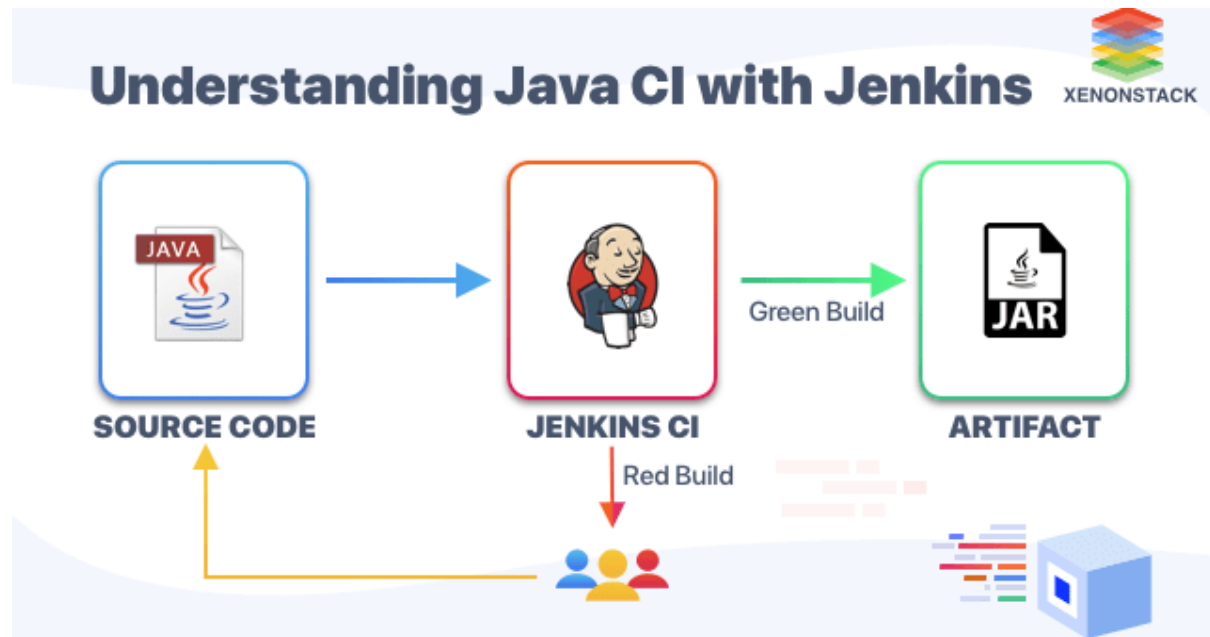




Jenkins

How to Build Java Projects in Jenkins :-

Jenkins allows us to automate the build and deployment process easily using Freestyle Projects. Follow the steps below to build a Java project:



Step-by-Step Process :-

1. Open Jenkins Dashboard

Launch Jenkins in your browser and log in using your credentials.

2. Create a New Freestyle Project

- Click on “New Item”
- Enter a project name
- Select “Freestyle Project”
- Click OK

3. Configure Source Code Management (SCM)

- Scroll to Source Code Management
- Select Git
- Enter your repository URL (GitHub/Local Git)

4. Add Build Steps

- Scroll to the Build section
- Click “Add Build Step”
- Select “Invoke top-level Maven targets” (if your Java project is Maven-based)
 - Enter Goals: clean install
OR
- Select “Execute Windows batch command”
 - Example Command:
 - javac YourFile.java
 - java YourFile

5. Post-Build Actions (Optional)

You can configure:

- Test reports
- Email notifications
- Deployment steps

6. Save & Build

Click Save and then Build Now from the left side menu.

You can check:

- Console Output → Build logs
- Workspace → Generated files (like .class / .jar)

Step 1: Access Jenkins and Log In



Sign in to Jenkins

Invalid username or password

Username

Password

☐ Keep me signed in

Sign in

Step 2 : Click the New Item link on the left-hand side of the Jenkins dashboard.



Step 3 : Enter the new project's name in the Enter an item name field and select the Freestyle project type. Click OK to continue.

New Item

Enter an item name

Java Project2

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK

Step 4 : Under the General tab, add a project description in the Description field.

General

Description

This is the Java Project.

Plain text [Preview](#)

☐ Discard old builds [?](#)

☐ GitHub project

☐ This project is parameterized [?](#)

☐ Throttle builds [?](#)

☐ Execute concurrent builds if necessary [?](#)

Advanced [v](#)

Source Code Management

Save

Apply

Step 5 : Under the Triggers tab & choose the the option as Build periodically.

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☒ Build periodically ?

Schedule ?

 No schedules so will never run

- ☐ GitHub hook trigger for GITScm polling ?
- ☐ Poll SCM ?

Step 6 : Under the Environment tab, Add the timestamps to the Console Output of the project.

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s) ?
- ☒ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☐ Terminate a build if it's stuck
- ☐ With Ant ?

Step 7 : Add Build the steps using the Execute windows batch command to build & run the java project.

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See [the list of available environment variables](#)

```
javac Addition.java  
java Addition
```

Advanced ▾

+ Add build step

Save

Apply

Step 8 : Click the Build Now link on the left-hand side of the new project page.



Jenkins / Java Project2

Status

Changes

Workspace

Build Now

Configure

Delete Project

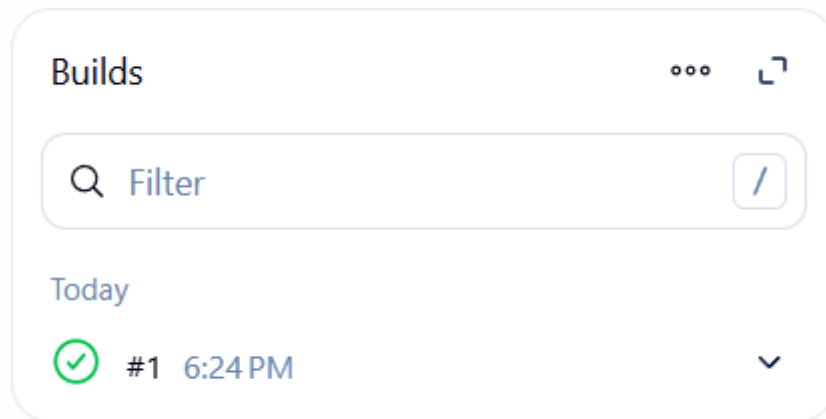
Rename

Java Project2

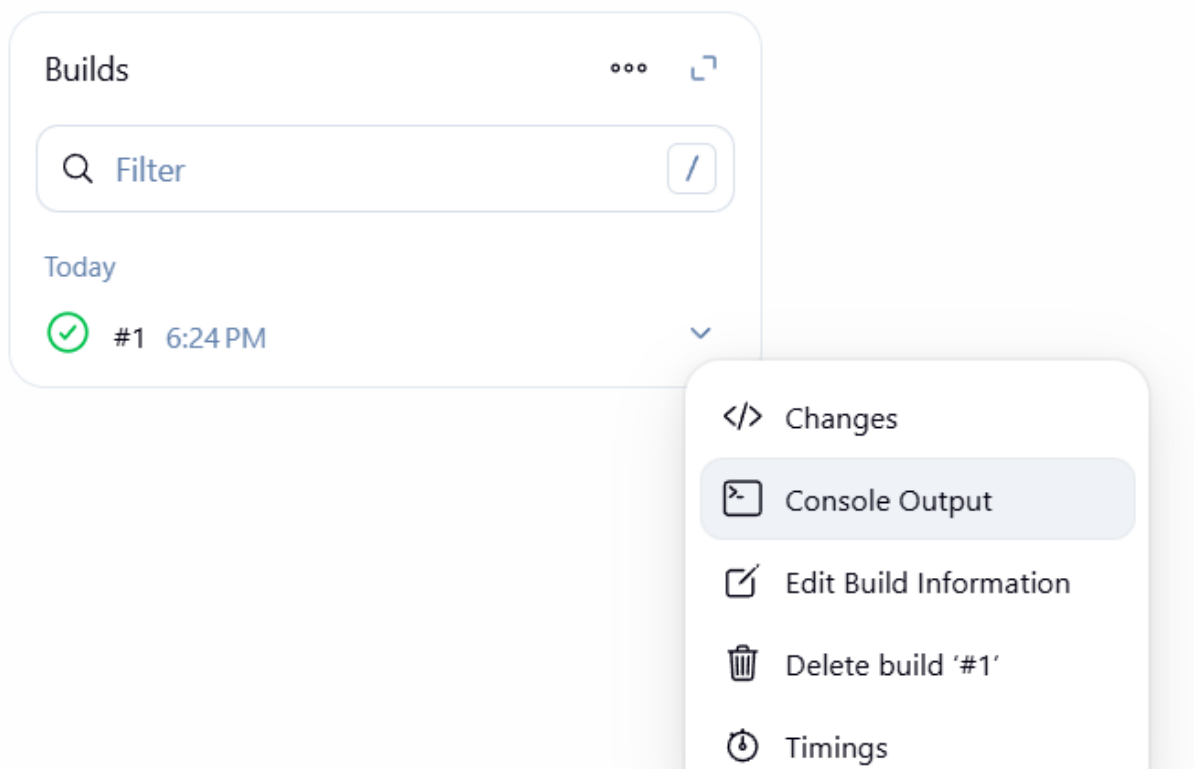
This is the Java Project.

Permalinks

Step 9 : Click the link to the latest project build in the Build History section.



Step 10 : Click the Console Output link on the left-hand side to display the output for the commands you entered.



Step 11 : The console output indicates that Jenkins is successfully executing the commands, displaying the current version of Java and Jenkins working directory.



Console Output

```
18:24:06 Started by user Sagar Haridas Kate
18:24:06 Running as SYSTEM
18:24:06 Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\Java Project2
18:24:06 [Java Project2] $ cmd /c call C:\WINDOWS\TEMP\jenkins11704467092942150207.bat
18:24:06
18:24:06 C:\ProgramData\Jenkins\.jenkins\workspace\Java Project2>javac Addition.java
18:24:08
18:24:08 C:\ProgramData\Jenkins\.jenkins\workspace\Java Project2>java Addition
18:24:08 The sum is: 15
18:24:08
18:24:08 C:\ProgramData\Jenkins\.jenkins\workspace\Java Project2>exit 0
18:24:08 Finished: SUCCESS
```